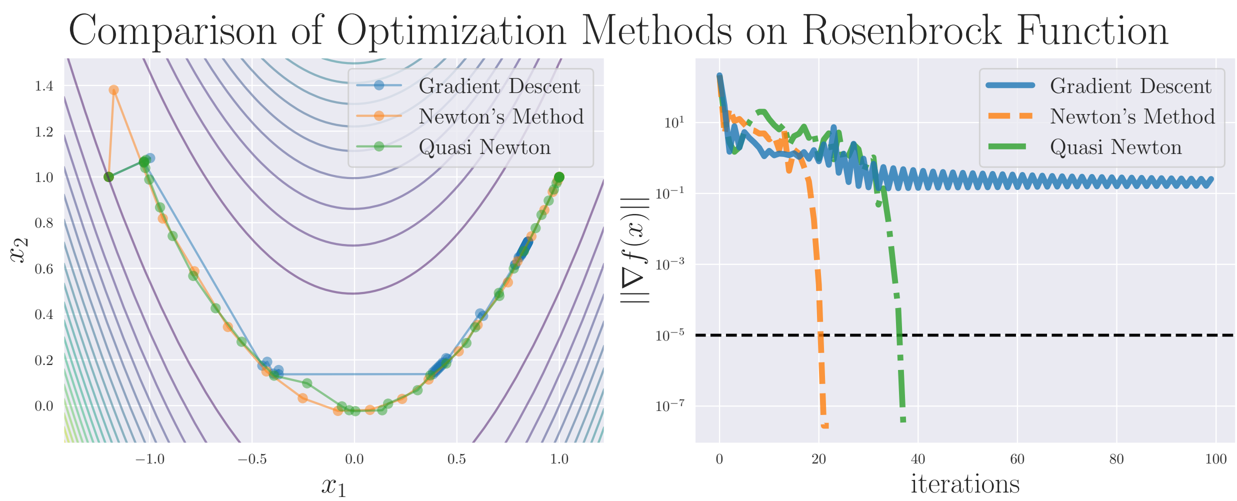

Figure 1: Comparison of Newton's method, quasi-Newton methods, and gradient descent.

1 Quasi-Newton Methods

In this section, we discuss quasi-Newton methods. Quasi-Newton methods are based on Newton's method but aim to reduce its primary drawback, namely the high computational cost of evaluating the Hessian matrix. Instead of using the true Hessian matrix $\nabla^2 f(x_k)$, an approximation matrix B_k is employed, thereby achieving fast convergence while keeping the computational cost low.

Figure 1 shows a comparison of Newton's method, quasi-Newton methods, and gradient descent. Gradient descent updates the iterate in the direction opposite to the gradient. Although its per-iteration computational cost is the lowest, its convergence is slow. Newton's method requires the fewest iterations, but it has the highest computational cost per step. Quasi-Newton methods lie between these two extremes and strike a balance between them. In particular, when the methods are compared in terms of total computation time rather than the number of iterations, quasi-Newton methods often exhibit the best performance.

Line-search-based quasi-Newton methods generate a sequence $\{x_k\}_k$ that converges to the optimal solution x^* as follows:

$$x_{k+1} = x_k - \alpha_k B_k^{-1} \nabla f(x_k) = x_k - \alpha_k H_k \nabla f(x_k).$$

Here, $\alpha_k > 0$ is the step size determined by line search, B_k is an approximation of the Hessian matrix $\nabla^2 f(x_k)$ at the point x_k, and $H_k := B_k^{-1}$ denotes its inverse. This update rule closely resembles that of Newton's method, which is why it is called a quasi-Newton method.

The core of quasi-Newton methods lies in how to update B_k at each iteration so that it approaches the true Hessian matrix. Figure 2 illustrates this concept. First, around the current point x_k, a quadratic approximation model of the objective function f is constructed using B_k. Next, this quadratic model is minimized to obtain the next point x_{k+1}. After obtaining x_{k+1}, the approximation matrix B_k is updated to B_{k+1} using the gradient information at x_k and x_{k+1}. This procedure is repeated until convergence, which constitutes the quasi-Newton method.

1.1 Secant Condition

Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be of class C^2. Suppose that a symmetric matrix B_k is given as an approximation of the Hessian matrix $\nabla^2 f(x_k)$. We consider updating the approximate Hessian to B_{k+1} for the next point x_{k+1}. Although there are infinitely many possible candidates for such a matrix, it is natural to impose symmetry on B_{k+1}, since the true Hessian matrix is symmetric. For the current point x_k and the next point x_{k+1}, we define the step and the gradient difference as follows:

$$s_k := x_{k+1} - x_k, \qquad y_k := \nabla f(x_{k+1}) - \nabla f(x_k).$$

In the following, we generally assume that $s_k \neq 0$ and $y_k^\top s_k \neq 0$. Note that the s in s_k stands for "step." Since the new approximation is constructed around x_{k+1}, we approximate $\nabla f(x_k)$ using the Hessian matrix $\nabla^2 f(x_{k+1})$ and its

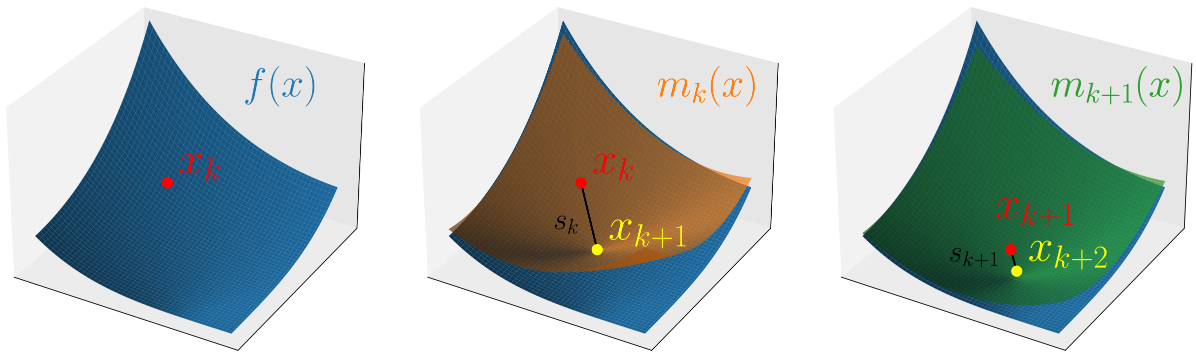

Figure 2: Conceptual illustration of quasi-Newton methods. (1) Given the objective function f (blue surface) and the current point x_k (red point). (2) Move to the minimizer x_{k+1} (yellow cross) of the quadratic model $m_k(x)$ (orange surface) constructed using the current approximate Hessian. (3) Update the quadratic model (green surface) and repeat this procedure.

approximation B_{k+1} as follows:

$$\begin{aligned}\nabla f(x_k) &\approx \nabla f(x_{k+1}) + \nabla^2 f(x_{k+1})(x_k - x_{k+1}) \\ &\approx \nabla f(x_{k+1}) + B_{k+1}(x_k - x_{k+1}).\end{aligned}$$

Requiring the above approximation to hold as an equality yields

$$B_{k+1}(x_{k+1} - x_k) = \nabla f(x_{k+1}) - \nabla f(x_k).$$

Equivalently, this can be rewritten as

$$B_{k+1} s_k = y_k.$$

This relation is called the secant condition or the quasi-Newton equation.

More precisely, the secant condition can also be justified as follows. Consider a quadratic approximation model around x_{k+1}:

$$m_{k+1}(x) = f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B_{k+1}(x - x_{k+1}).$$

By construction, regardless of B_{k+1}, this model satisfies

$$\begin{cases} m_{k+1}(x_{k+1}) = f(x_{k+1}), \\ \nabla m_{k+1}(x_{k+1}) = \nabla f(x_{k+1}). \end{cases}$$

In addition, the secant condition ensures that the model satisfies

$$\begin{aligned}\nabla m_{k+1}(x_k) &= \nabla f(x_{k+1}) - B_{k+1}(x_{k+1} - x_k) && \text{(definition of the model)} \\ &= \nabla f(x_{k+1}) - (\nabla f(x_{k+1}) - \nabla f(x_k)) && \text{(secant condition)} \\ &= \nabla f(x_k).\end{aligned}$$

This implies that the secant condition ensures that the quadratic model $m_{k+1}(x)$ accurately reflects the known information at both x_{k+1} and x_k, except for the previous function value $f(x_k)$.